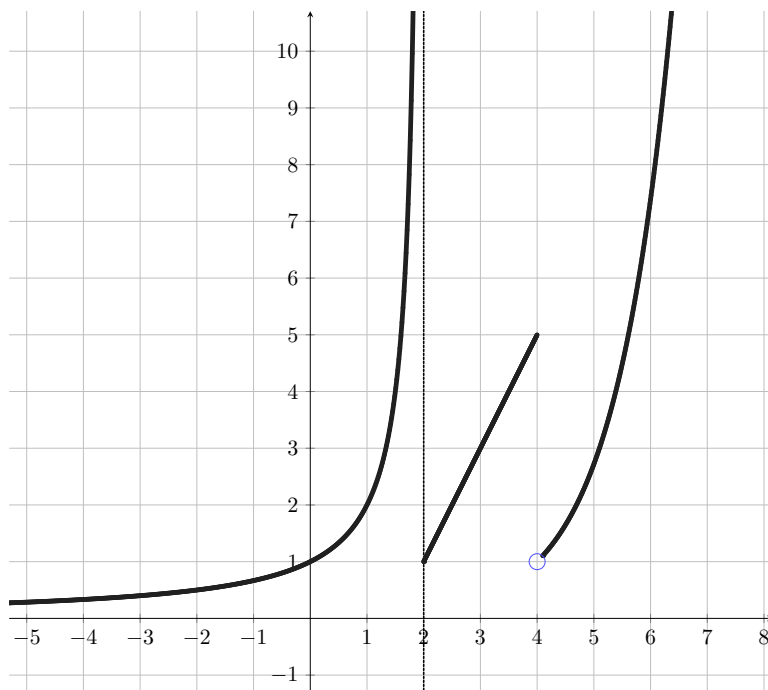


KUWAIT UNIVERSITY
Department of Mathematics, Calculus I (Math 101)
First Exam Solutions

1. [3 + 6 + 6 = 15 pts.] The graph of the function f is given. Use it to answer each of the following:



- (a) Find $\lim_{x \rightarrow 4} f(x)$, if it exists. Justify your answer.

We have $\lim_{x \rightarrow 4^+} f(x) = 1$ and $\lim_{x \rightarrow 4^-} f(x) = 5$. Thus $\lim_{x \rightarrow 4} f(x)$ Does not exist.

- (b) Find all vertical and horizontal asymptotes of the function f . Justify your answer.

We have $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow 2^-} f(x) = \infty$. Thus $y = 0$ and $x = 2$ are the horizontal and vertical asymptotes of the given function, respectively.

- (c) Find all points of discontinuity and determine the type of each one. Justify your answer.

The function has infinite discontinuity at $x = 2$ since $\lim_{x \rightarrow 2^-} f(x) = \infty$ and has jump discontinuity at $x = 4$ since $\lim_{x \rightarrow 4^-} f(x) = 5$ and $\lim_{x \rightarrow 4^+} f(x) = 1$.

2. [15 pts.] Find the domain, the range and a formula for the inverse of the function $f(x) = \frac{2 \ln x - 3}{3 \ln x - 6}$.

We have $3 \ln x - 6 = 0$ and this is equivalent to $x = e^2$. Thus the domain of f is $(0, \infty) \setminus \{e^2\}$.

Now to find the inverse, we have $y = \frac{2 \ln x - 3}{3 \ln x - 6}$. Now if we solve for x in terms of y , we obtain

$x = e^{\frac{6y-3}{3y-2}}$. Therefore, the inverse of f is given by $f^{-1}(x) = e^{\frac{6x-3}{3x-2}}$. Finally, the range of f is equal to the domain of f^{-1} that is $\mathbb{R} \setminus \{\frac{2}{3}\}$.

3. [10 pts.] Let $f(x) = \begin{cases} \frac{x^2 - 3x + 2}{x^2 - 1}, & \text{if } x > 1, \\ ax + 2, & \text{if } x \leq 1. \end{cases}$

Find the value of the constant a for which the function f is continuous at $x = 1$.

The function f is continuous at $x = 1$ if $f(1) = \lim_{x \rightarrow 1} f(x)$. This is equivalent to

$$a + 2 = \lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 1} = \lim_{x \rightarrow 1^+} \frac{(x-2)(x-1)}{(x+1)(x-1)} = \lim_{x \rightarrow 1^+} \frac{x-2}{x+1} = \frac{-1}{2}.$$

The last statement implies that $a = \frac{-5}{2}$.

4. [10 pts.] Evaluate the limit $\lim_{x \rightarrow 3} \frac{x-3}{2-\sqrt{x+1}}$, if it exists.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x-3}{2-\sqrt{x+1}} &= \lim_{x \rightarrow 3} \frac{x-3}{2-\sqrt{x+1}} \times \frac{2+\sqrt{x+1}}{2+\sqrt{x+1}} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(2+\sqrt{x+1})}{4-(x+1)} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(2+\sqrt{x+1})}{3-x} \\ &= \lim_{x \rightarrow 3} -(2+\sqrt{x+1}) = -4. \end{aligned}$$

5. [15 pts.] Given that $2\sin(x-1) \leq (x-1)f(x) \leq 2x-2$. Use Squeeze Theorem to find the value of $\lim_{x \rightarrow 1^+} f(x)$, if it exists.

We divide the inequality by $x-1$ with the assumption that $x > 1$ to obtain

$$\frac{2\sin(x-1)}{x-1} \leq f(x) \leq \frac{2x-2}{x-1}.$$

Now we have $\lim_{x \rightarrow 1^+} \frac{2\sin(x-1)}{x-1} = \lim_{u \rightarrow 0^+} \frac{2\sin u}{u} = 2 = \lim_{x \rightarrow 1^+} 2 = \lim_{x \rightarrow 1^+} \frac{2(x-1)}{x-1} = \lim_{x \rightarrow 1^+} \frac{2x-2}{x-1}.$

Therefore by Squeeze Theorem, we obtain $\lim_{x \rightarrow 1^+} f(x) = 2$.

6. [10 pts.] Use the definition of the derivative to find the derivative of the function $f(x) = \frac{2}{x+1} + 1$.

We have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2}{x+h+1} + 1 - \frac{2}{x+1} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{2}{x+h+1} - \frac{2}{x+1}}{h} = \lim_{h \rightarrow 0} \frac{\frac{2x+2-2x-2h-2}{(x+1)(x+h+1)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2h}{h(x+1)(x+h+1)} = \frac{-2}{(x+1)^2}. \end{aligned}$$

7. [15 pts.] Find the point(s) on the graph of $f(x) = \frac{x^2 - 3}{x + 2}$ at which the tangent line is horizontal.

The tangent line is horizontal whenever $f'(x) = 0$ that is when $\frac{(x+2)(2x) - (x^2-3)(1)}{(x+2)^2} = 0$ using quotient rule or simply $x^2 + 4x + 3 = 0$ which implies that $x = -1, -3$. Thus the graph of f has horizontal tangent line at the points $(-1, f(-1)) = (-1, -2), (-3, f(-3)) = (-3, -6)$.

8. [10 pts.] Use the Intermediate Value Theorem to show that the equation $\tan x + 2x - 2 = 0$ has at least one real root.

The function $f(x) = \tan x + 2x - 2$ is continuous on $\left[0, \frac{\pi}{4}\right]$ since it is the sum of a trigonometric function and a linear function with domain that contains the given interval $\left[0, \frac{\pi}{4}\right]$. Also, we have $f(0) = -2 < 0 < \frac{\pi}{2} - 1 = f\left(\frac{\pi}{4}\right)$. Therefore by the Intermediate Value Theorem, there exists $c \in \left(0, \frac{\pi}{4}\right)$ such that $f(c) = 0$ or simply $\tan c + 2c - 2 = 0$ which implies that c is a real root of the given equation.